

Soumil Gupta

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Computer Architect

EDUCATION

Carnegie Mellon University

Expected May 2027

Master of Science in Electrical & Computer Engineering

University of Illinois, Urbana-Champaign

May 2025

Bachelor of Science in Computer Engineering | GPA: 3.8/4.0 | Honors | Dean's List

Relevant Coursework:

- **Hardware:** Computer Architecture, Digital Systems, VLSI System Design, IC Device Theory and Fabrication, Semiconductor Electronics, Digital Signal Processing, Analog Signal Processing
- **Software:** Operating Systems, Applied Parallel Programming, Communication Networks, Algorithms & Models of Computation, Data Structures & Algorithms, Computer Systems & Programming
- **AI/ML:** Artificial Intelligence, Applied Machine Learning, Deep Learning for CV, IoT & Cognitive Computing

TECHNICAL SKILLS

Languages: C++, C, C#, Python, Java, x86 Assembly

EDA/RTL DV: SystemVerilog, SV Coverage, Intel Quartus, AMD Xilinx Vivado, Synopsys (DC, VCS, Spyglass, Verdi), Verilator, Chipyard, ModelSim/Questasim, Cadence Virtuoso, Cadence Innovus, Cocoth, Chisel, HLS, PyRTL, Veryl, Gem5, UVM

Interfaces & Test: AXI, UART, SPI, I2C, I2S, CAN, JTAG, ATPG, BIST, ATE, TCP/IP

GPU Systems Tools: CUDA, Nsight Systems, Nsight Compute, Compute Sanitizer, cuDNN, cuBLAS, TensorRT

ML & Scripting Libraries: Tensorflow, Keras, PyTorch, NumPy, SciPy, Scikit-Learn, OpenCV, Pandas

Tools/Environments: Linux, GDB, Valgrind, Bash/Shell, TCL, Vim, Git, Jenkins, Docker, QEMU, Jupyter, LaTeX

Other: ROS 1/2, RViz, Gazebo, RPi, Arduino, Matlab, Simulink, Unity, Eagle CAD, Autodesk Inventor, SolidWorks

WORK EXPERIENCE

CPU Performance Architect Intern | Samsung

Sept 2025 - Dec 2025

- Worked with senior engineers **developing new line of Samsung mobile SoCs for on-device AI** based on **ARM architecture**.
- **Modeled industry-leading IP** for CPU microarchitecture components (branch predictors, prefetchers, load-store queues, etc.) to **analyze power, performance, area (PPA) tradeoffs**.
- **Simulated, debugged, and validated CPU features using Gem5** and C++ simulation, emulation, and analysis tools.

Undergraduate Teaching Assistant | Computer Systems & Programming Course, UIUC

Aug 2024 - May 2025

- Led weekly office hours with 5-10 students per session, guiding students in developing algorithms and debugging with GDB.
- **Taught C & C++ programming fundamentals**, file/device I/O, memory, signals & interrupt handling, and concurrency.

RESEARCH EXPERIENCE

Researcher | Future Architecture & Systems Technology (FAST) Lab, UIUC

Apr 2024 - May 2025

- **Write, verify, layout, and tapeout** proof-of-concept SoC with Berkeley RISC-V Cores, [MESA Controller](#), & CGRA.
- **Developed custom ISA** for MESA controller to map CPU programs with any **CGRA Hardware Accelerator** via SDFGs.
- **Built custom hardware verification framework** complete with golden model to compare DUT v. gold execution traces.
- **Verified SystemVerilog, Chisel modules** on AXI units, register files, etc., with ModelSim, Verilator, and Vivado software.

Researcher | Autonomous & Unmanned Vehicle Systems Laboratory (AUVSL), UIUC

Dec 2022 - May 2025

- Led a team to design and implement novel outdoor robot localization solution to solve robot autonomy in GPS-denied areas by leveraging Ultrawide-Band Antennas. **Wrote Python & C++ packages**, deployable on Clearpath robot platforms.
- **Developed PyTorch scripts** to tune Adaptive Neural Fuzzy Inference System parameters, implemented SHFAF localization algorithm that improved localization accuracy by 82% to within 10 cm.
- **Customized C firmware** and Utilized SPI communication protocol with Decawave UWB Sensors.
- [Co-authored IEEE paper](#) on localization methods. Presented updates, deliverables to CERL, U.S. Army Corps of Engineers.

HARDWARE SYSTEMS PROJECT EXPERIENCE

RISC-V 32IM N-Way Superscalar Out-of-order Explicit Register Renaming CPU in SystemVerilog

May 2024

- Designed & verified from scratch an out-of-order N-way parameterizable processor for RISC-V 32-bit ISA with M-extension, achieving IPC of 0.51 on standard benchmark taking 21.7 mW of power at clock frequency 325 MHz.
- Implemented speculative branching with branch prediction overriding, & perceptron and Gshare branch predictors.
- Implemented cache features including next-line & stride prefetchers, post-commit store buffer with write coalescing.
- Integrated Synopsys IPs including sequential divider, and wrote Dadda advanced multiplier and shift-add multiplier.
- Wrote full processor in SystemVerilog, Simulated with Synopsys VCS, Debugged with Synopsys DC, Verdi, Spike, RVFI, Lint, Functional Coverage, Used Python to generate test programs, Built toolchains with bash, GNU make.

RISC-V 32I Bitsliced CPU in Cadence Virtuoso & Innovus

Apr 2025

- Authored FreePDK 45nm standard library (logic, muxes, flip-flops) with schematic, layout, and Liberty timing views.
- Manually laid out in Virtuoso a complete 32-bit RISC-V RV32I pipeline using a bitsliced methodology in a hierarchical library, complete with ALU, write-low/read-high register file, barrel-shifter, and cascaded comparator.
- Verified extracted netlist with SystemVerilog harness, passing reference, and self-written RV32I assembly suites.
- Wrote Tcl scripts for Innovus to make 2nd processor in addition to the manual layout automating the layout process including floorplanning, placement, routing, & DRC/LVS checks, & minimize area for the entire processor.

IC Fabrication - Course-Based Project at UIUC

Dec 2024

- Independently processed a silicon wafer from start to finish in an industrial cleanroom environment, performing every step—from RCA cleaning and oxidation to photolithography, etching, doping, and final metal lift-off.
- Utilized a five-mask sequence to fabricate complex devices (MOSFETs, BJTs, diodes, logic gates).

FPGA-Accelerated DNN for Autonomous Traffic Sign Recognition

Apr 2025

- Designed a hardware-accelerated Deep Neural Network for AMD PYNQ-Z2 FPGA using AMD Xilinx Vivado HLS.
- Implemented convolution, activation, and pooling layers with HLS pragmas to optimize dataflow and pipelining.
- Integrated the accelerator via Vivado AXI IP blocks and automated image feeding through a Python script.
- Leveraged ScaleHLS framework & hardware-software co-design, balancing accuracy with FPGA resource limits.
- Employed TensorFlow to train & deploy models on GPUs, Raspberry Pi, Google Edge TPU for benchmarking.

FPGA Implementation of Digital Audio Workstation (DAW) with Audio Processor

May 2023

- Independently created DAW (like GarageBand) for creating music on Intel MAX10 FPGA that records, crops, applies audio effects, and mixes four stereo tracks on SDRAM, drives 640x480 VGA interface, and responds to USB-keyboard input. Audio effects include 6 dB gain, low-pass FIR equalization, and sample-accurate stereo-mixing.
- Engineered custom audio pipeline in C & SystemVerilog: SGTL5000 audio codec (ADC/DAC) converts analog-digital audio; I2S shifters send 32-bit stereo samples, and embedded-C I2C firmware programs codec registers.
- Researched documentation to incorporate Intel FPGA peripherals, NIOS II, and Unit-Tested functionalities.

SOFTWARE PROJECT EXPERIENCE

Custom UNIX Operating System

Dec 2023

- Wrote custom UNIX OS for emulated single-core Intel Pentium IV Processor from scratch using C & x86 Assembly.
- Implemented a paging-only virtual memory system, writable file system, software context switching, terminal switching, and hardware interrupts and exception handling.
- Developed device drivers for keyboard, mouse, real-time clock, & built interactive shell for executing system calls.
- Used advanced GDB techniques to efficiently debug and resolve system-level issues.
- Emulated and tested the OS in a QEMU virtual environment, ensuring safe and efficient debugging.

TCP Protocol Implementation

Oct 2024

- Implemented a C-based transport protocol to emulate TCP-like behavior using UDP sockets.
- Designed sender/receiver functions to ensure accurate, high-efficiency transfer, handling packet drops & reordering.
- Setup Docker-based testing environment & Utilized Wireshark, tcpdump, & libpcap to analyze network performance.

AI/ML PROJECT EXPERIENCE

LeNet-5 Implementation in CUDA

May 2025

- Designed and implemented a version of LeNet-5 CNN to run on a cluster of Nvidia A40 GPUs.
- Built convolution layers with fused feature-map unrolling, tiled GEMM, and output permutation into one Tensor-Core (WMMA) kernel, maximizing throughput.
- Profiled hot spots with Nsight Systems and Nsight Compute; validated memory using compute-sanitizer and ensured bit-exact parity against a CPU reference instrumented with gprof.
- Overlapped PCIe moves and compute via dual async CUDA streams (req0), hiding latency to 256-image batch sizes.

Deep Reinforcement Learning for Atari Breakout

Apr 2024

- Implemented DQN and Double DQN agents with Python, PyTorch, and OpenAI Gym to master Atari Breakout.
- Developed a recurrent state mechanism to maintain historical context, enhancing decision-making during gameplay.
- Tuned hyperparameters and executed training on a GPU-based Linux cluster to achieve target mean scores.
- Incorporated LSTM modules to further improve temporal dependency handling and performance.

RESEARCH PUBLICATIONS

1. **S. Gupta**¹, "Implementation of a CPU Controller for Dynamic Binary Translation," May 2025 ([Accepted to IDEALS](#)).
2. M. Juston¹, **S. Gupta**¹, et al., "Robust Error State Sage-Husa Adaptive Kalman Filter for UWB Localization," IEEE Sensors Journal, vol. 25, no. 9, pp. 16034–16049, Mar 2025, [doi: 10.1109/JSEN.2025.3549315](#).